

# AI基礎設施融資

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## AI融資格局顯示Anthropic算力需求面臨三路競逐

(Bloomberg Intelligence) -- Anthropic的算力建設是跨越融資與晶片結構的三方角力，而非雙線之爭。AWS依托Trainium和Inferentia，仍是Anthropic的主要雲服務與訓練合作夥伴；同時，谷歌和博通正借助私募信貸工具、殘值支持及場地擔保，通過基於Fluidstack的項目推進張量處理器(TPU)。據報導，輝達支持的Volta與比特小鹿達成Anthropic相關協議，表明GPU路線正在新興雲服務商層面作出回應。關鍵問題在於，哪種架構能整合晶片、信貸以及稀缺的已供電場地資源，同時避免業主困於單一生態。(08/12/26)

### 1. 算力SPV分離TPU硬體風險

算力特殊目的實體(SPV)將谷歌-博通路線轉變為一套融資邏輯，而非單純的晶片供應敘事。《金融時報》報導稱，該實體斥資約350億美元購入了約1吉瓦的TPU硬體，並將其租賃給Anthropic；Apollo和Blackstone作為債務融資的核心出資方，博通則將為優先檔提供殘值支持。

隨著AWS、谷歌與輝達均承接Anthropic需求，這一結構明確了風險歸屬。場地業主與Fluidstack未必持有其中規模最大的硬體資產。在這條路線中，TPU可由私募信貸出租方持有，谷歌可為數據中心場地提供擔保，博通則吸收殘值風險。(08/12/26)

AI風險架構將晶片與場地分離

| AI infrastructure layer   | Primary party           | What sits on balance sheet / contract                       | BI read-through                                                                              |
|---------------------------|-------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| End demand                | Anthropic               | Long-term compute payment commitments                       | AI lab drives utilization, but spreads demand across AWS, Google and Nvidia lanes            |
| TPU hardware owner        | Compute SPV             | TPU pods financed with private-credit debt                  | Hardware risk can sit outside Fluidstack and site landlords                                  |
| Residual-value support    | Broadcom                | Support for senior debt if TPU resale value falls short     | Broadcom helps make specialized TPUs financeable outside Google Cloud                        |
| Physical-site backstop    | Google                  | Data-center lease guarantees / step-in support              | Google supports the real estate side of the TPU buildout                                     |
| Site / cloud intermediary | Fluidstack              | Lease and AI-cloud operating role in Fluidstack-based sites | Fluidstack may control customer/site orchestration without owning the largest hardware layer |
| Power, land, shell        | Landlords / site owners | Construction debt, powered shells and delivery obligations  | Public and private landlords still carry site-execution risk                                 |

Source: Bloomberg Intelligence

Bloomberg Intelligence BI

### 2. 博通擔保提升TPU的可融資性

博通的殘值支持使谷歌TPU得以在谷歌雲體系之外具備融資可行性：若這類專用硬體無法以足夠價值轉售，博通將為私募信貸機構提供保護。據《金融時報》報導，在初始規模350億美元的TPU融資工具中，博通為其中約300億美元的優先檔債務提供支撐，若Anthropic停止付款且硬體銷售無法覆蓋優先債權人本金，博通將補足差額。

由此，博通的角色從設計合作夥伴，延伸至供應商融資支持方。相較於部署更為廣泛的輝達GPU，TPU與谷歌生態的綁定更為緊密，因此債權人需要更強的擔保，才願意將其視為抵押品。博通的信貸支持既保護了其定制晶片業務，也鞏固了其在Anthropic算力建設中的地位。該SPV讓谷歌得以為場地提供擔保，而Anthropic實現規模化擴張的同時無需承擔資本支出。(08/12/26)

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殘值支持為TPU債務架構托底

| Financing layer    | Party                           | Role                                                                            | Risk read-through                                                            |
|--------------------|---------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| TPU supply         | Google / Broadcom               | TPU pods move into a dedicated financing vehicle                                | Custom silicon is separated from Google Cloud's balance sheet                |
| Hardware owner     | Compute SPV                     | Owns TPU pods and leases them to Anthropic                                      | Private-credit vehicle carries the hardware asset                            |
| Customer cash flow | Anthropic                       | Pays lease obligations tied to TPU usage                                        | AI-lab demand services the financing                                         |
| Senior debt        | Apollo / Blackstone-led lenders | Fund the SPV through debt tranches                                              | Lenders rely on lease cash flow, collateral and support                      |
| Residual backstop  | Broadcom                        | Supports senior debt if Anthropic stops paying and TPU resale value falls short | Broadcom makes specialized TPUs financeable despite thinner resale liquidity |

Source: Bloomberg Intelligence

Bloomberg IntelligenceBI

3. Volta挑戰谷歌圍繞Anthropic的TPU路線

面對谷歌為Anthropic打造的TPU路線，輝達以Volta作為現有戰略基礎上的進一步回應。據傳的Anthropic算力協議，加之比特小鹿官方公佈的Volta Tydal租約，表明輝達支持的新興雲服務商算力能夠直接爭奪前沿實驗室的需求，而無需完全依賴傳統雲服務巨頭承擔GPU工作負載。

戰略意義並不單純在於新增一處輝達算力場地。Volta整合了GPU供給、資本形成以及與AI實驗室的關係，比特小鹿則提供具備供電條件的算力容量。這在功能上與谷歌路線如出一轍，儘管結構有所不同：谷歌與博通借助TPU、私募信貸、場地擔保以及Fluidstack關聯項目，將業務擴展至谷歌雲之外。(08/12/26)

兩條融資路線角逐AI需求

| Question                    | Google-Broadcom TPU lane                                       | Volta-Bitdeer GPU lane                                                                   |
|-----------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Who provides chips?         | Google TPUs co-developed with Broadcom                         | Nvidia GPUs                                                                              |
| Who intermediates capacity? | Fluidstack-based site structures                               | Volta                                                                                    |
| Who provides the site?      | Cipher, Hut 8, TeraWulf and other power-first landlords        | Bitdeer, also a power-first former crypto-site operator                                  |
| How is risk supported?      | Google site guarantees; Broadcom residual support for TPU debt | \$1.3 billion expected letters-of-credit backstop for Volta obligations                  |
| Who anchors demand?         | Anthropic-linked TPU buildout                                  | Leading AI lab; Bloomberg reported Anthropic                                             |
| Why it matters              | Extends TPUs beyond Google Cloud                               | Shows the GPU lane can contest the same AI-lab demand through neoclouds and scarce power |

Source: Bloomberg Intelligence

Bloomberg IntelligenceBI

4. 比特小鹿將挪威電力轉化為AI租金

比特小鹿的Tydal租約將挪威電力轉化為長期的AI基礎設施收入，使其成為與Cipher、Hut 8和TeraWulf相比，電力資源向AI收入轉化路徑更清晰的可比公司。為期16年的Volta租約涵蓋121兆瓦的IT負荷，對應133兆瓦的總容量，專為某領先AI實驗室配置輝達GPU，戴爾擔任技術供應商。

該租約具有47億美元的基礎租期價值，若續約八年則可能達到80億美元，以及預計13億美元的信用證擔保和預計約90%的淨經營收入利潤率，共同勾勒出從服務於加密貨幣挖礦的電力資產向AI基礎設施業主模式轉型的經濟路徑。比特小鹿保留園區100%所有權，且未發行認股權證，使得其結構與谷歌支持的Fluidstack租賃模式截然不同。

(08/12/26)

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比特小鹿憑Tydal租約躋身AI業主行列

| Metric                         | Bitdeer Tydal disclosure                                    | BI read-through                                  |
|--------------------------------|-------------------------------------------------------------|--------------------------------------------------|
| Capacity                       | 121 IT MW supported by 133 gross MW                         | Meaningful AI site scale                         |
| Lease value                    | \$4.7 billion over 16 years                                 | Converts Norway power into long-duration AI rent |
| Extension option               | Potential \$8.0 billion over 24 years                       | Extends visibility if renewed                    |
| Credit support                 | About \$1.3 billion letters of credit anticipated           | Volta obligations get institutional backstop     |
| Economics                      | About \$202 per kW per month; about 90% expected NOI margin | Shows attractive site-level lease math           |
| Ownership                      | Bitdeer keeps 100% of Tydal; no warrants issued             | Differs from Google-backed Fluidstack structures |
| Source: Bloomberg Intelligence |                                                             | Bloomberg Intelligence BI                        |

### 5. GPU路線通過運營商為AI提供資金

輝達路線仍需要巨額融資，但並不需要谷歌TPU融資工具那樣依賴博通提供殘值支持。谷歌-博通模式通過結構化信用支持安排，使專用TPU硬體在谷歌雲體系之外具備可融資可行性；而GPU路線更多依靠CoreWeave、IREN、Volta和比特小鹿等運營商及場地平台籌措資金，以客戶合約、已供電場地及可重複部署的GPU抵押資產作為債務支撐。

這一區別對於風險承銷至關重要。谷歌正借助私募信貸、殘值支持與場地擔保構建谷歌外部TPU生態，而輝達生態則可能依托更深厚的二級市場需求、運營商資產負債表以及債權人對GPU資產的熟悉度。這場競賽的結果取決於哪種架構能規模化整合電力、硬體與信貸資源。(08/12/26)

TPU與GPU路線拆分融資風險

| Financing issue                | Google-Broadcom TPU lane                                           | Nvidia GPU lane                                                                              | BI read-through                                                         |
|--------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Hardware collateral            | Specialized TPUs financed through Compute SPV                      | Reusable Nvidia GPUs financed through operators                                              | GPUs may be easier collateral; TPUs need more engineered credit support |
| Vendor support                 | Broadcom residual-value support backs senior TPU debt              | Nvidia support more often comes through supply, ecosystem ties, equity or warrants           | Less need for Nvidia to guarantee residual value directly               |
| Site support                   | Google backstops physical data-center leases                       | Site risk sits with operators or landlords such as Volta, Bitdeer, CoreWeave, IREN or others | Google lane separates chip and site guarantees more explicitly          |
| Customer cash flow             | Anthropic lease payments support TPU hardware financing            | AI-lab or hyperscaler contracts support operator debt and site leases                        | Both lanes depend on large customers continuing to absorb capacity      |
| Main risk                      | Can private credit underwrite less-liquid custom silicon at scale? | Can operators fund GPU fleets and keep utilization high?                                     | Financing architecture is becoming as important as chip performance     |
| Source: Bloomberg Intelligence |                                                                    | Bloomberg Intelligence BI                                                                    |                                                                         |

### 6. 機構業主可更長期保持中立

機構數據中心業主則帶來一個重要的限定條件：並非所有AI業主都會捲入谷歌、輝達或AWS陣營。Vantage、Aligned、Compass和Stack更能保持硬體中立，因為機構發起人、規模化資產組合以及超大規模租戶基礎，為其提供了比依賴單一錨定協議的電力優先型平台更為深厚的融資渠道。Equinix與Digital Realty還提供數據中心托管和互聯服務模式，但它們的共同優勢都在於租戶中立的基礎設施。

這種中立性具有戰略價值。比特小鹿、Cipher、Hut 8和TeraWulf可通過簽署與特定晶片生態或新興雲服務商掛鉤的錨定合約，加快項目推進速度，但此類協議可能收窄未來的租賃選擇範圍。房地產層面討論的重點在於：規模、租戶信用及中立性是否比綁定單一架構能帶來更穩健的經濟效益。(08/12/26)

資本模式驅動AI中立性

| Model                          | Examples                             | Core offering             | AI exposure          |
|--------------------------------|--------------------------------------|---------------------------|----------------------|
| Institutional                  | Vantage; Aligned; Compass; Stack     | Campuses                  | More neutral         |
| Colo                           | Equinix; Digital Realty              | Interconnection           | More neutral         |
| Power-first                    | Bitdeer; Cipher; Hut 8; TeraWulf     | Power sites               | Anchor-contract pull |
| Neocloud                       | Fluidstack; Volta; CoreWeave; Nebius | Compute                   | Chip-linked          |
| Source: Bloomberg Intelligence |                                      | Bloomberg Intelligence BI |                      |

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7. 風險分層追蹤工具勾勒AI債務擴張分析框架

AI基礎設施債務的層次過於複雜，難以用總量數字概括。一筆與Anthropic相關的建設項目，即可能涉及算力SPV硬體債務、博通殘值支持、谷歌場地擔保、Fluidstack或Volta租賃義務、業主建設債務，以及來自AI實驗室的長期客戶付款承諾。風險評估需要一個追蹤工具，按場地、晶片結構、交易對手方及信貸支持，區分硬債務、類債務擔保、租賃承諾與客戶合約。

這一框架有助於揭示，投資者究竟是在為可重複利用的已供電容量承擔信用風險，還是圍繞單一客戶、單一架構的履約義務鏈，後者承擔的壓力可能沿晶片、業主及私募信貸層層傳導。(08/12/26)

AI風險存在於不同層次

| Risk layer           | Example                       | Disclosed scale            | What it means                       |
|----------------------|-------------------------------|----------------------------|-------------------------------------|
| Hardware debt        | Compute SPV TPU vehicle       | \$35B                      | Private credit funds chip ownership |
| Vendor backstop      | Broadcom residual support     | About \$30B                | Protects senior TPU lenders         |
| Site guarantee       | Google data-center backstops  | Up to \$44B if leases fail | Supports physical site leases       |
| Lease credit support | Volta Tydal letters of credit | About \$1.3B               | Backstops tenant obligations        |
| Site revenue         | Bitdeer Tydal base lease      | \$4.7B over 16 years       | Turns power into AI rent            |
| Landlord debt        | Construction financing        | Varies by site             | Funds power and shell delivery      |

Source: Bloomberg Intelligence

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